

INFORMATION TECHNOLOGY INSTITUTE

Full-Stack Web Development — Laravel & Vue.js

# Attendance & Grading Platform

Project Requirements Specification

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Capstone Project | Team of 4–5 | Full Stack  
Version 1.0

# 1. Project Overview

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This document specifies the requirements for an Attendance & Grading Platform serving a single ITI branch that runs multiple training tracks (for example, Web Development and Mobile Development) in parallel. Each track runs one active cohort at a time.

The platform is a full-stack web application. The backend is a Laravel REST API and the frontend is a Vue 3 single-page application that consumes it. The two are developed and graded with equal weight.

The system manages the full operational loop of a cohort: opening the cohort, assigning staff, scheduling teaching engagements, recording per-session attendance, grading students, surfacing analytics to each role, and forwarding billable instructor hours to central accounting.

## 1.1 Goals

- Provide a single source of truth for attendance and grades across all tracks in the branch.
- Give each role a dashboard scoped to exactly what they are allowed to see.
- Automate grade computation, including normalization and a standalone attendance ledger.
- Automate billable-hour calculation from the session schedule and forward it to central accounting.
- Flag at-risk students early, before the cohort ends.

## 1.2 Out of Scope

- Multi-branch operation. The system serves one branch only.
- Public self-registration. All accounts are provisioned top-down (see Section 9).
- Payment processing. The platform produces billing data and forwards it to central accounting; it does not pay anyone.

## 1.3 Glossary

| Term              | Meaning                                                                           |
|-------------------|-----------------------------------------------------------------------------------|
| Branch            | The single ITI location the system serves.                                        |
| Track             | A discipline within the branch, e.g. Web or Mobile.                               |
| Cohort            | One intake of a track (e.g. “Web — Intake 45”). One active cohort per track.      |
| Course            | A graded subject inside a cohort, scored out of 100.                              |
| Engagement        | A teaching booking (lecture, lab, or business session) assigned to an instructor. |
| Lab group         | A subset of a cohort (~15 students) assigned to one lab instructor.               |
| Attendance ledger | A standalone per-student point balance starting at 250.                           |

## 2. Roles & Access Model

The system distinguishes a person's access profile (what they can see and do) from their engagements (sessions they are booked to teach) and their compensation type (how they are paid). These are deliberately decoupled.

### Key design principle

A role is not the same as a teaching seat. A person with an administrative role can also hold teaching engagements across different tracks. For example, a Track Admin for Web may also be booked to deliver lab sessions for Mobile. Their access stays tied to their role; their pay gains an hourly component for the sessions they deliver.

### 2.1 Roles

| Role           | Scope                            | Core responsibilities                                                                                                                                                                                            |
|----------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Branch Manager | Whole branch                     | Creates cohorts; assigns Track Admins; views executive analytics and the branch billing rollup.                                                                                                                  |
| Track Admin    | Their track's full cohort roster | Configures courses, grade weights, lab groups; schedules engagements and assigns instructors; approves excuses; enters final exam/project grades; overrides instructor evaluations; posts announcements anytime. |
| Instructor     | Only their assigned lab group(s) | Takes attendance; grades their own group; adds student tags/notes; posts announcements during their active engagement window only.                                                                               |
| Student        | Their own data only              | Views grades and attendance; reads announcements; submits assignments; requests excuses.                                                                                                                         |

### 2.2 Compensation Types

| Type                   | Formula                                                 |
|------------------------|---------------------------------------------------------|
| External instructor    | Pure hourly. Paid only for delivered session hours.     |
| Track Admin (internal) | Fixed salary + (delivered session hours × hourly rate). |

*All delivered hours are calculated automatically from the schedule and forwarded to central accounting (see Section 7).*

### 2.3 Visibility Rules

**ACC-1** The Branch Manager shall see branch-wide aggregated analytics with drill-down into any track and cohort.

**ACC-2** A Track Admin shall see the full roster and all grades for their own track's active cohort at any time.

**ACC-3** An Instructor shall see only the students in the lab group(s) they are assigned to. An instructor assigned to groups 1 and 2 shall not see group 3.

**ACC-4** A Student shall see only their own grades, attendance, and submissions. No peer data is visible to students.

**ACC-5** Student tags/notes shall be visible to every person who grades that student, internal or external (see Section 6.5).

### 3. Cohort Lifecycle

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A cohort moves through the following stages. Each stage is driven by a specific role.

1. **Open cohort.** The Branch Manager creates a cohort for a track and assigns one or more Track Admins.
2. **Configure.** The Track Admin sets up courses and their grade weights, creates lab groups, schedules engagements, and assigns/hires instructors.
3. **Deliver.** Instructors deliver sessions, record attendance, and grade their lab groups. The Track Admin enters final exam/project grades.
4. **Participate.** Students attend (scan in/out), submit deliverables, request excuses, and view their grades and announcements.
5. **Roll up.** Billable hours flow to central accounting; performance metrics flow to the Branch Manager dashboard.

**LC-1** A track shall have at most one active cohort at any time.

**LC-2** Only the Branch Manager shall create cohorts and assign Track Admins.

## 4. Engagements & Scheduling

An engagement is a teaching booking with a type, a date range, scheduled session hours, and an assigned instructor. The Track Admin creates engagements; the system uses their scheduled hours to compute billing.

### 4.1 Engagement Types

| Type             | Attendance        | Deliverable                                | Notes                                                                                                                         |
|------------------|-------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Lecture          | Required          | None                                       | Standard taught session.                                                                                                      |
| Lab              | Required          | Worth 10 points; MANDATORY for the student | Track splits into 2–3 lab groups of ~15; each group's instructor grades only that group. Late penalty applies (see 4.2).      |
| Business session | Tracked per track | None                                       | Cross-track: students from multiple tracks (e.g. Web + Mobile) may attend the same session. Attendance is recorded per track. |

### 4.2 Lab Deliverable Penalty

**ENG-1** A lab deliverable shall be worth 10 points and shall be optional for the student.

**ENG-2** Each full day late shall deduct 25% of the deliverable's total. After 4 full days late the deliverable score reaches 0.

**Example:**

A deliverable scored 10/10 submitted 2 days late:  $10 - (2 \times 25\% \times 10) = 5$  points.

The same deliverable submitted 4 days late:  $10 - (4 \times 25\% \times 10) = 0$  points.

### 4.3 Scheduling Rules

**ENG-3** Each engagement shall record its type, assigned instructor, date range, and scheduled hours per session.

**ENG-4** A person may hold multiple engagements across different tracks within their active window.

**ENG-5** An instructor's account access shall be limited to their engagement's date range (see Section 9 on account expiry).

## 5. Attendance

Attendance is recorded per session as two timestamps: an “arrived at” scan-in and a “left at” scan-out. The scanner interface is intentionally minimal: scan, show arrived-at, scan again, show left-at.

### 5.1 Check-in Mechanism

**ATT-1** The system shall support QR-code check-in and check-out, recording a timestamp for each.

**ATT-2** The scanner interface shall be a single, dead-simple screen optimized for fast repeated scanning.

**ATT-3** Business-session attendance shall be recorded per track even when multiple tracks attend the same session.

**BONUS** NFC check-in via the Web NFC API. Web NFC is supported only on Chrome for Android, so any NFC implementation must fall back to QR on unsupported devices. This is optional and earns bonus marks.

### 5.2 Attendance Ledger

Each student has one standalone attendance ledger that spans the whole program. It is a points balance, not a percentage, and it is added to the grand total as-is (see Section 6.1).

| Event                                  | Effect on ledger |
|----------------------------------------|------------------|
| Starting balance                       | 250 points       |
| Unexcused absence (any session type)   | -25 points       |
| Excused absence (Track Admin approved) | -5 points        |

**ATT-4** Every student's ledger shall start at 250 points.

**ATT-5** An unexcused absence shall deduct 25 points; an approved excused absence shall deduct 5 points.

**ATT-6** Missing any session type, including business sessions, shall affect the same single ledger.

### 5.3 Excuse Workflow

An absence becomes “excused” only after a Track Admin approves the student's request. The workflow is a simple state machine.

**requested** → **approved** | **rejected**

**EXC-1** A Student shall submit an excuse request with a reason and an optional attachment.

**EXC-2** An attachment shall be no larger than 1 MB and limited to PDF or image formats. The system shall validate size and type on upload.

**EXC-3** A Track Admin shall approve or reject each request. Approval changes the deduction from 25 to 5; rejection leaves it at 25.



## 6. Grading

### 6.1 Grade Architecture

Grading is two-tier. The attendance ledger stands on its own and is added directly to the grand total. Every course is scored out of 100 and contributes its own score.

#### Grand total formula

Grand Total = Attendance Ledger (out of 250) + Sum of Course Scores (each out of 100)

The attendance ledger is NOT folded into any course. It is a separate line item added as-is.

### 6.2 Within a Course (out of 100)

Each course's 100 points are split by weight across its components —

1. lab deliverables
2. final exam/project. *The Track Admin* sets these weights once per cohort.

**GRD-1** A course shall be scored out of 100, split by weight across its graded components.

**GRD-2** Component weights shall be set once per cohort by the Track Admin.

### 6.3 Normalization

A component's raw maximum rarely equals the points it is worth in the course. The system normalizes the student's raw score onto the component's weight:

$$\text{componentScore} = (\text{rawScore} \div \text{rawMax}) \times \text{componentWeight}$$

#### Example

The assignments component is worth 40 points of the course's 100.

Assignments were graded out of a raw maximum of 70.

A student scoring 67/70 raw earns:  $(67 \div 70) \times 40 = 38.3$  points toward the course.

**GRD-3** The system shall normalize each component's raw score onto its configured weight using the formula above.

### 6.4 Lab Groups, Final Grade & Overrides

A track of roughly 45 students splits into 2–3 lab groups of about 15. Each group's instructor grades only their own students. As an example, all assignments of a course might be worth 40 of the 100, while the final exam/project carries the rest.

**GRD-4** Each lab instructor shall grade only the students in their assigned group.

**GRD-5** The Track Admin shall enter the final exam/project grade for each course.

**GRD-6** A Track Admin shall be able to override an instructor's evaluation. An override shall require a mandatory note explaining it, and the original value shall be retained for audit.

## 6.5 Student Tags & Notes

Tags and notes accumulate on a student across courses and follow them through the cohort. They give later instructors context from earlier ones.

**GRD-7** The system shall support both predefined tags (e.g. “uses AI”, “Cheating”, “loves extra work”) and free-text notes on a student.

**GRD-8** Tags and notes shall be accumulative across courses and visible to everyone who grades that student, internal or external.

## 7. Billing & Central Accounting

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Billing is derived from the schedule, not entered by hand. The system totals each person's delivered session hours and forwards the result to central accounting.

**BIL-1** The system shall auto-calculate billable hours per person from their scheduled and delivered sessions.

**BIL-2** External instructors shall be billed purely on delivered hours; internal Track Admins shall be billed for hours on top of their fixed salary.

**BIL-3** The system shall produce a consolidated billing rollup forwarded to central accounting.

**BIL-4** The Branch Manager shall see the billing rollup, including the split between internal and external hours, within their dashboard.

## 8. Student Portal & Announcements

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### 8.1 Student Portal

**POR-1** A Student shall view their attendance ledger and their per-course score breakdown by component.

**POR-2** A Student shall view their own attendance record session by session.

**POR-3** A Student shall view their own progress over time. No peer comparison shall be shown.

**POR-4** A Student shall submit assignments either as a URL (e.g. a repository or drive link) or as a direct file upload. Both methods shall be supported.

**POR-5** A Student shall submit and track excuse requests (see Section 5.3).

### 8.2 Announcements

Announcements are article-style posts shown in the student's feed.

**ANN-1** A Track Admin shall post announcements to their cohort at any time.

**ANN-2** An Instructor shall post announcements only during their active engagement window.

**ANN-3** Announcements shall support article-style content and appear in the student feed.

## 9. Accounts, Authentication & Security

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**SEC-1** There shall be no public self-registration. All accounts are provisioned top-down: Branch Manager → Track Admins → instructors and students.

**SEC-2** Every account shall carry an expiry date. Expired accounts shall be unable to log in (e.g. an external instructor's account expires at the end of their contract; student accounts expire at cohort end).

**SEC-3** The API shall enforce role- and scope-based authorization on every endpoint, matching the visibility rules in Section 2.3. Authorization shall be enforced server-side, never only in the Vue client.

**SEC-4** File uploads (excuse attachments, assignment files) shall be validated for size and type and stored safely minio(s3 combatable storage is a bonus).

## 10. Analytics & Dashboards

Each role gets a dashboard scoped to its visibility. The two highlighted analytics below are where teams can differentiate and earn bonus marks.

### 10.1 At-Risk Rule

A student is flagged at-risk when either condition holds:

- Attendance ledger falls below 150 points, OR
- Any single course grade falls below 60.

**ANL-1** The system shall flag at-risk students using the rule above and surface them in the relevant dashboards.

### 10.2 Dashboards by Role

| Role           | Dashboard contents                                                                                                                                                                                                    |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Branch Manager | Cross-track comparison (attendance %, average grade, pass/dropout rate); cohort attendance trend over time; billing/cost view (total hours, internal vs external split, cost-per-student); drill-down track → cohort. |
| Track Admin    | Cohort grade distribution; lab-group grader-consistency check; attendance per session and per student; deliverable submission status; tag-flagged and at-risk students.                                               |
| Instructor     | Their group's grade distribution; submission tracker (late/missing); their own delivered hours.                                                                                                                       |
| Student        | Own grade breakdown by component; own attendance record; own progress over time.                                                                                                                                      |

### 10.3 Highlighted (Bonus) Analytics

**Grader-consistency analysis.** Surface whether one lab instructor grades systematically harder or more leniently than another — for example, group 3's mean sitting several points below groups 1 and 2 before normalization. This helps the Track Admin spot inconsistent grading.

**At-risk detection.** Beyond the flat rule in 10.1, teams may add early-warning signals such as attendance trend slope or a sudden drop between courses.

## 11. Technical Requirements & Deliverables

### 11.1 Stack

- **Backend:** Laravel REST API, with database migrations, seeders, and validation.
- **Frontend:** Vue 3 single-page application consuming the API.
- **Auth:** Token-based authentication with server-side, role-scoped authorization.

*Laravel and Vue are weighted equally in evaluation.*

### 11.2 Expected Deliverables

6. A working full-stack application implementing all mandatory requirements in this document.
7. Database schema with migrations and seed data sufficient to demonstrate every role and flow.
8. API documentation (e.g. a Postman collection or OpenAPI spec).
9. A short README covering setup, seeded test accounts for each role, and any assumptions made.
10. A live demo walking through each role's experience end to end.

### 11.3 Team

This is a team project for 4–5 members. Teams are expected to divide work across the backend API, the Vue client, and the cross-cutting concerns (auth, analytics, attendance scanning), and to integrate continuously rather than at the end.

### 11.4 Mandatory vs. Bonus

| Bonus item                              | Where        |
|-----------------------------------------|--------------|
| NFC check-in (Web NFC) with QR fallback | Section 5.1  |
| Grader-consistency analytics            | Section 10.3 |
| Advanced at-risk early-warning signals  | Section 10.3 |

#### A note on requirement IDs

Every requirement in this document is tagged (ACC-, ATT-, GRD-, and so on). Use these IDs in your commit messages and branch names (BONUS)